

NORTHWIND SALES DASHBOARD REPORT

1. INTRODUCTION

The Northwind Sales Dashboard is a Power BI project created to analyse sales performance using the Northwind dataset. The dashboard helps to understand important information such as total sales, product categories, customer details and overall sales performance.

Power BI was used to clean and transform the data, create calculated measures, and present the information through interactive charts, KPI cards, slicers and other visualisations. The dashboard makes it easier to identify sales trends and compare different categories.

2. OBJECTIVES

The main objectives of the Northwind Sales Dashboard are:

- To analyse the overall sales performance.
- To identify the best-performing product categories.
- To display important sales information using KPI cards.
- To transform and clean the dataset using Power Query.
- To create measures using DAX for sales analysis.
- To present data using suitable charts and visualisations.
- To provide interactive filtering using slicers.
- To analyse detailed information using drill-through.
- To identify important sales trends and key findings.
- To make the sales data easier to understand and interpret.

3. DATA TRANSFORMATION USING POWER QUERY

Power Query was used to prepare the Northwind dataset before creating the dashboard. The data was checked and transformed so that it could be used properly for analysis.

The main data transformation activities included:

- Checking the available tables and columns.
- Removing unnecessary data where required.
- Changing data types correctly.
- Cleaning and organising the data.
- Renaming columns where necessary.
- Handling missing or incorrect values.
- Creating relationships between relevant tables.
- Loading the cleaned data into Power BI.

These transformations helped to improve the quality and accuracy of the data used in the dashboard.

4. MEASURES USED

DAX measures were created in Power BI to calculate important sales values. Measures help to analyse the data dynamically when filters and slicers are applied.

The important measures used in the dashboard include:

Measure	Purpose
Total Sales	Shows the overall sales amount
Total Quantity	Shows the total quantity of products sold
Average Sales	Shows the average sales value
Total Orders	Shows the number of orders
Sales	Helps to understand changes in

Growth	sales
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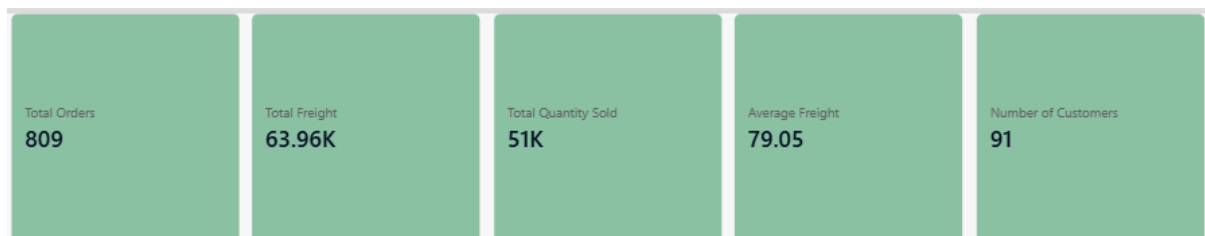
5. KPI CARDS

KPI cards are used to display important business information in a simple format. They provide a quick summary of the main performance indicators.

The dashboard uses KPI cards to show values such as:

- Total Sales
- Total Orders
- Total Quantity
- Average Sales
- Other important sales indicators

KPI cards help the user understand the overall performance without checking every chart individually.



6. VISUALISATIONS USED

6.1 ORDERS BY MONTH:

I created a Line Chart using the Order Date month and Total Orders. This visual helps me understand how the number of orders changes from month to month. It makes it easier to identify periods with relatively higher or lower order activity.

Data Labels:

Data labels can be used to display the exact number of orders on the chart.

Tooltip:

When I hover over a particular month, the tooltip provides the month and corresponding Total Orders value.

6.2 SALES BY PRODUCT CATEGORY:

I created a Column Chart using Category Name and Total Sales. This visual helps me compare the sales generated by different product categories and understand the contribution of each category to overall sales.

Data Labels:

The data labels display the exact sales value for each category.

Tooltip:

The tooltip provides:

- Category Name
- Total Sales
- Total Quantity Sold

This allows me to obtain additional numerical information without overcrowding the main chart.

6.3 ORDERS BY EMPLOYEE:

I created a Bar Chart using Employee Name and Total Orders.

This visual helps me compare the number of orders associated with different employees.

Data Labels:

The exact Total Orders value can be displayed beside each bar.

Tooltip:

When I hover over an employee, the tooltip displays the Employee Name and Total Orders.

6.4 CUSTOMERS BY COUNTRY:

I created a visual to analyse the number of customers across different countries.

The Country field is used as the category and Number of Customers is used as the value.

This visual helps me understand the geographical distribution of the Northwind customer base.

Data Labels:

The data labels display the number of customers for each country.

Tooltip:

The tooltip provides:

- Country
- Number of Customers
- Total Sales

6.5 PRODUCT-WISE ANALYSIS:

I created a detailed table containing product-level information.

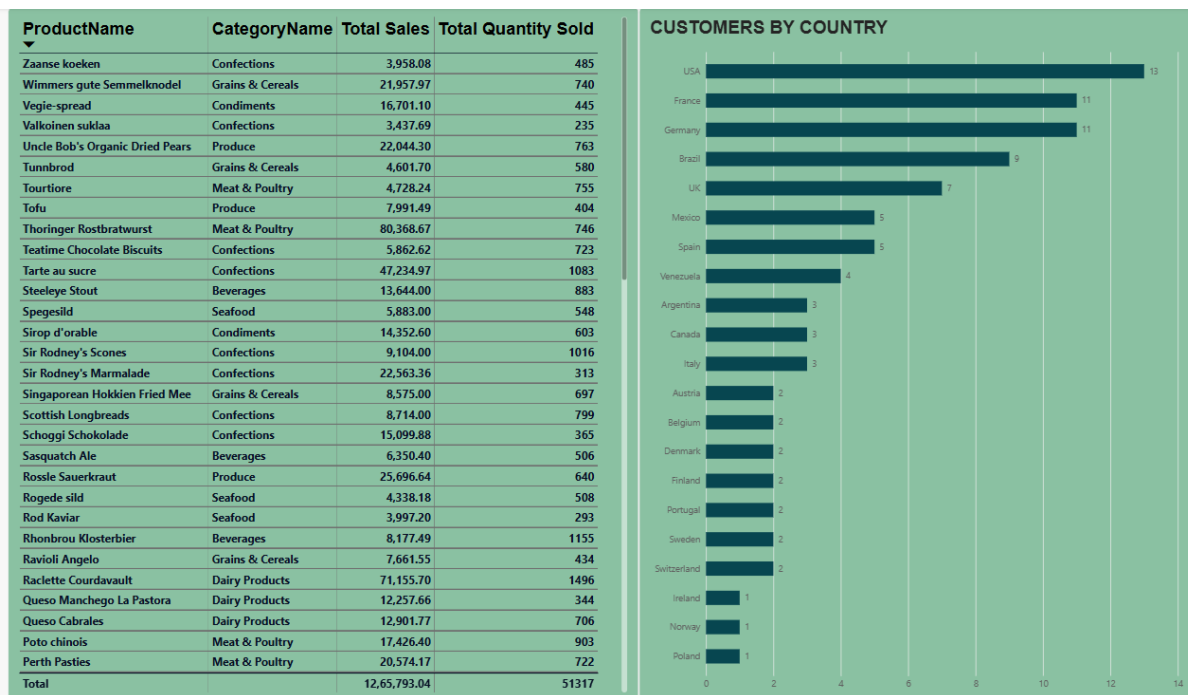
The table includes:

- Product Name
- Category Name
- Total Sales
- Total Quantity Sold
- Total Orders

This table allows me to compare individual products and understand their sales and quantity performance.

It also helps identify products that contribute significantly to the overall business performance.





7. CATEGORY NAME STANDARDISATION

Data transformation was performed to maintain consistency in category names.

The category information was checked during the Power Query transformation stage to ensure that the category names were consistent throughout the dataset.

The standardised category information is then used throughout the dashboard.

This helps maintain consistency in:

- Category slicer
- Category-related charts
- Product table
- Tooltip
- Category analysis
- Drill-through analysis

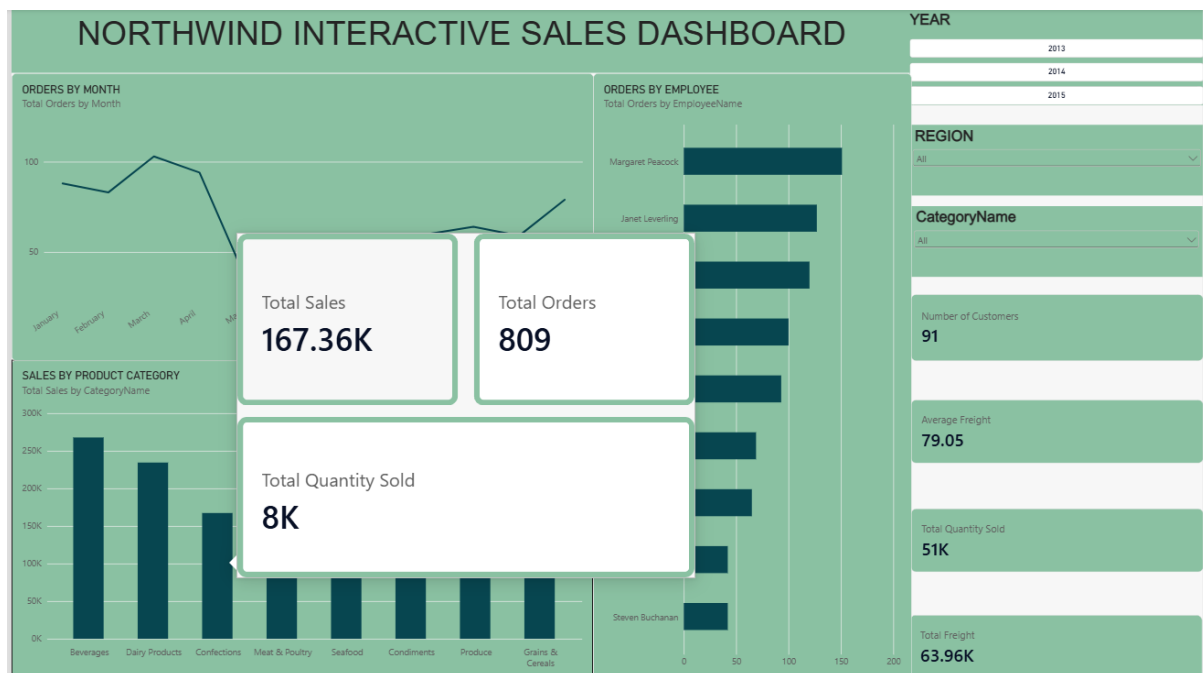
8. TOOLTIPS

I used tooltips to provide additional information when the user moves the cursor over a visual.

The main tooltip information includes:

Visual	Tooltip Information
Orders by Month	Month + Total Orders
Sales by Category	Category Name + Total Sales
Orders by Employee	Employee Name + Total Orders
Customers by Country	Country + Number of Customers
Product Analysis	Product + Sales + Quantity

Tooltips help keep the main dashboard clean while still allowing users to access detailed numerical information.



9. DATA LABELS

I used data labels in selected charts to display exact values directly on the visuals.

For example, data labels can display:

- Total Sales for each category
- Total Orders for each employee
- Number of Customers for each country
- Total Orders for monthly analysis
- Total Quantity Sold for selected products

Data labels make it easier to understand the exact numerical values without depending only on the visual axis or tooltip.

10. SLICERS AND INTERACTIVITY

I used slicers to make the dashboard interactive.

The main filtering fields include:

- Country
- Year
- Category
- Product

When a particular value is selected, the related visuals and KPI cards automatically update based on the selected filter.

This allows the user to analyse the Northwind data from different perspectives and compare specific sections of the business.

11. DRILL-THROUGH ANALYSIS

I also added a Drill-through feature to the Northwind dashboard.

The purpose of drill-through is to move from a summary visual to a separate detailed page based on the selected data point.

For example, when a user selects a particular product or category from the main dashboard and uses the Drill-through option, Power BI can take the user to the detailed analysis page.

The detailed page provides information such as:

Selected Product or Category:

- Total Sales
- Total Orders
- Total Quantity Sold
- Average Order Value
- Related product information

This feature is useful because the main dashboard provides a highlevel summary, while the drill-through page provides deeper information about the selected item.

Benefits of Drill-through:

- Provides detailed analysis without overcrowding the main dashboard.
- Allows users to investigate a selected product or category.
- Maintains the selected filter context.
- Connects summary-level analysis with detailed analysis.
- Makes the dashboard more interactive.

I also included a Back button on the detailed page so that the user can return to the main dashboard after completing the drill-through analysis.



12. DASHBOARD PAGES

The Northwind Power BI report contains the following pages:

1. SALES OVERVIEW

Provides the main summary of sales and order performance.

2. PRODUCT & CUSTOMER ANALYSIS

Provides detailed analysis of products and customers.

3. INTERACTIVE SALES DASHBOARD

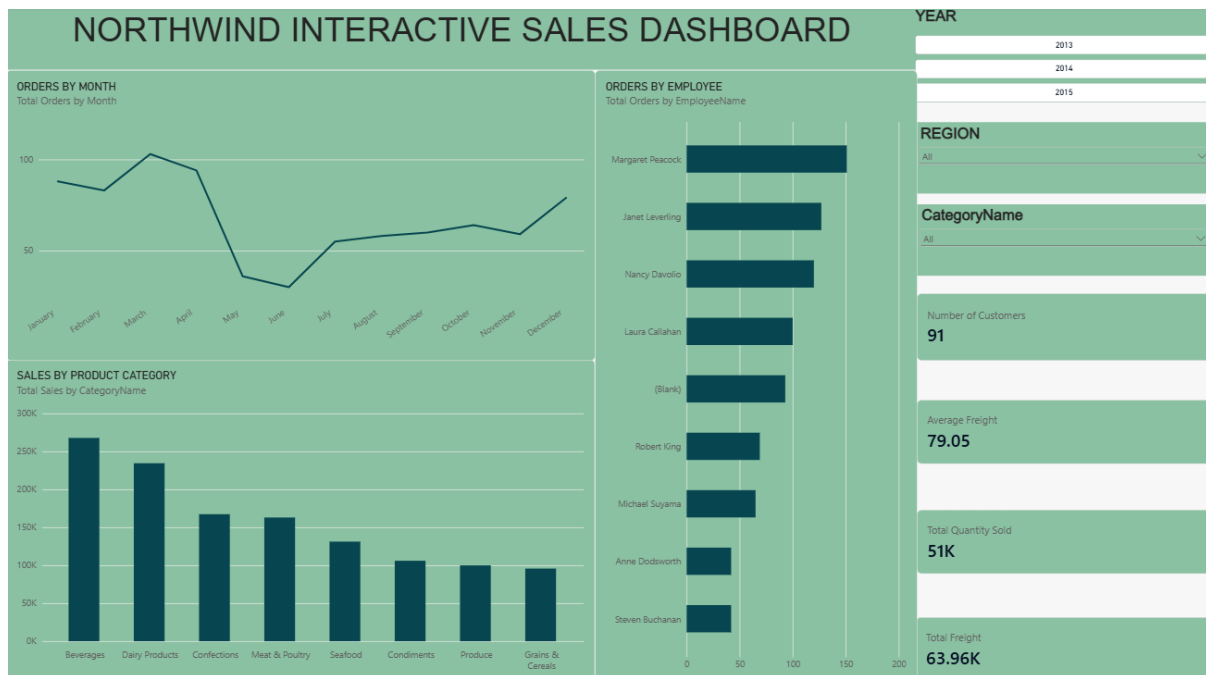
Provides interactive KPI cards, charts, slicers, tooltips, and other analysis.

4. CATEGORY TOOLTIP

Provides additional information related to category-level analysis.

5. CATEGORY DETAILS

Provides detailed information through the drill-through functionality.



13. KEY FINDINGS

The dashboard analysis provided the following key findings:

- Overall sales performance was analysed using the Total Sales KPI.
- Overall order activity was measured using the Total Orders KPI.
- The dashboard shows the Total Quantity Sold across the dataset.
- Total Freight and Average Freight provide information about freight-related performance.
- Customer distribution can be analysed across different countries.
- Beverages records the highest sales among the product categories, followed by Dairy Products.
- Employee-wise order activity can be compared using the Orders by Employee chart.

- Monthly order activity changes throughout the year, with variations between different months.
- Product-level analysis provides information about Product Name, Category Name, Total Sales and Total Quantity Sold.
- Tooltips provide additional information for different charts without overcrowding the main dashboard.
- Slicers allow the data to be analysed based on selected filters such as country, year, category and product.
- Drill-through provides detailed information about selected products or categories.
- Power Query helps in cleaning, transforming and preparing the dataset before analysis.
- DAX measures support the calculation and analysis of important business values.

14. CONCLUSION

The Northwind Sales Dashboard provides a clear and interactive view of sales, orders, products and customer information. The use of KPI cards, charts, tables, slicers and other interactive features makes the data easier to understand and analyse.

Overall, the dashboard helps identify sales trends, compare product categories, analyse employee-wise orders and understand customer and product details. It provides a simple and organised way to convert the available data into meaningful business information.